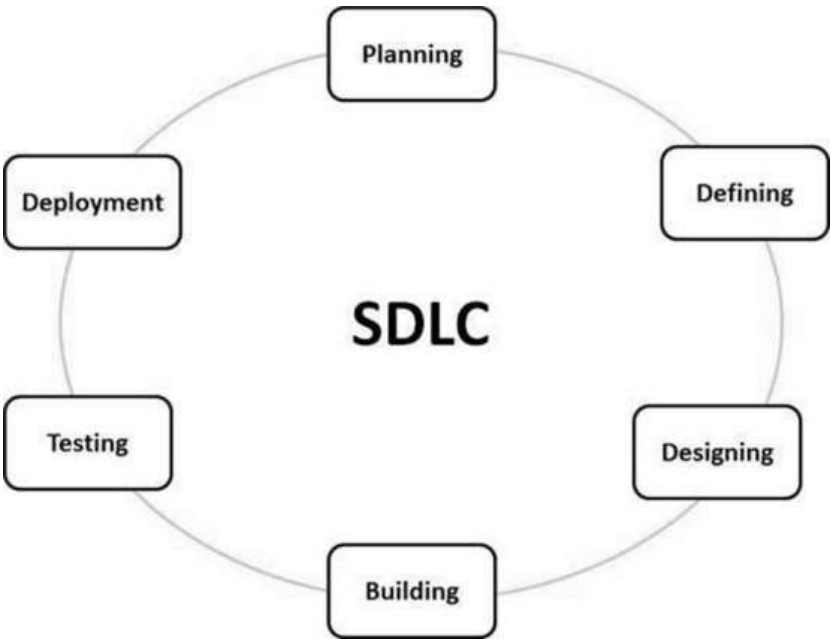
SDLC

Software Development Life Cycle

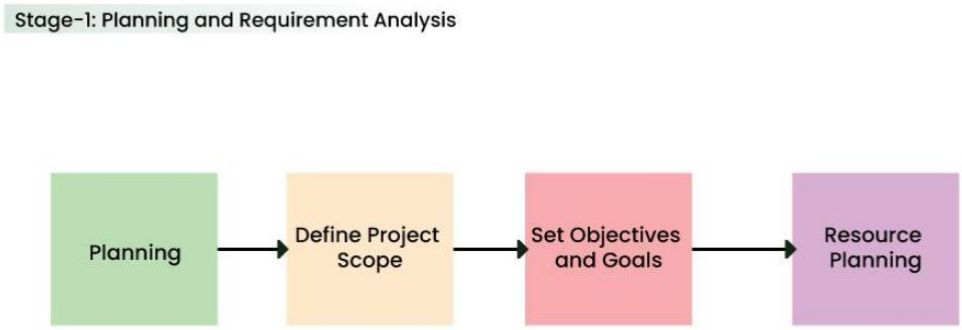
- SDLC is a process followed for a software project, within a software organization.
- It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software.
- The life cycle defines a methodology for improving the quality of software and the overall development process.
- The following figure is a graphical representation of the various stages of a typical SDLC.

SDLC


Planning
Defining
Designing
Building
Testing
Deployment
SDLC

SDLC Stages:

- Stage 1: Planning and Requirement Analysis



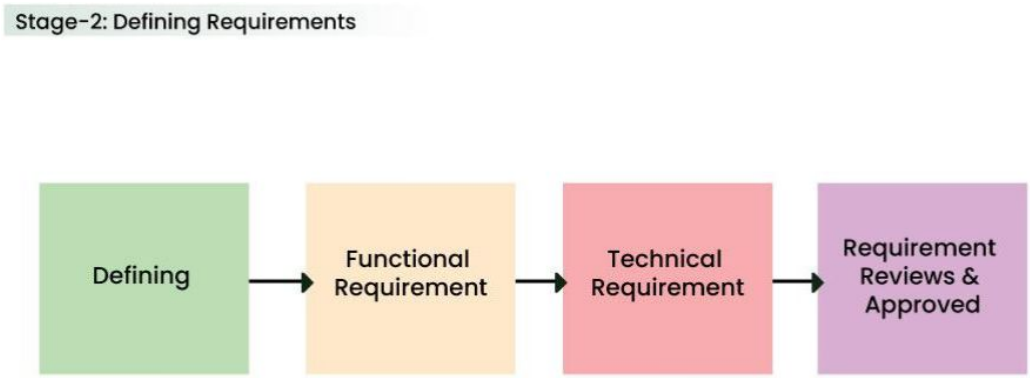
Stage 1: Planning and Requirement Analysis

- Requirement analysis is the most important and fundamental stage in SDLC.
- It is performed by the senior members of the team with inputs from the customer, the sales department, market surveys and domain experts in the industry.
- This information is then used to plan the basic project approach and to conduct product feasibility study in the economical, operational and technical areas.

Stage 1: Planning and Requirement Analysis

- Planning for the quality assurance requirements and identification of the risks associated with the project is also done in the planning stage.
- The outcome of the technical feasibility study is to define the various technical approaches that can be followed to implement the project successfully with minimum risks.
- Activities: Feasibility studies (technical, operational, economic), resource allocation, project scheduling, and cost estimation.
- Output: Project Plan, Feasibility Report.
- Key Players: Senior Engineers, Project Managers, Stakeholders.

Stage 2: Defining Requirements

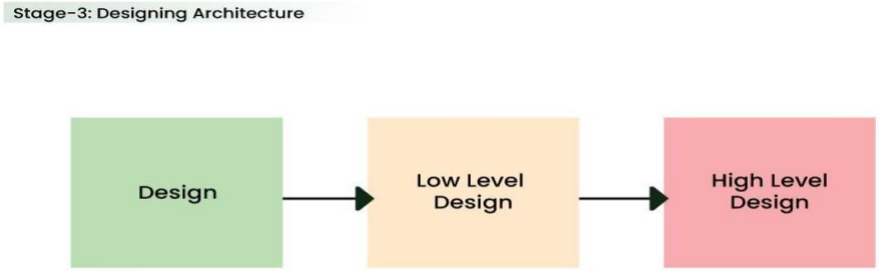


Stage 2: Defining Requirements

- Once the requirement analysis is done the next step is to clearly define and document the product requirements and get them approved from the customer or the market analysts.
- This is done through an SRS (Software Requirement Specification) document which consists of all the product requirements to be designed and developed during the project life cycle.
- Activities: Gathering detailed functional and non-functional requirements from customers.
- Output: SRS Document (The "Bible" for the development team).
- Key Players: Business Analysts (BAs), Product Owners

Stage 3: Designing the Product Architecture

- A design approach clearly defines all the architectural modules of the product along with its communication and data flow representation with the external and third party modules (if any). The internal design of all the modules of the proposed architecture should be clearly defined with the minutest of the details in DDS(Design Document Specification)

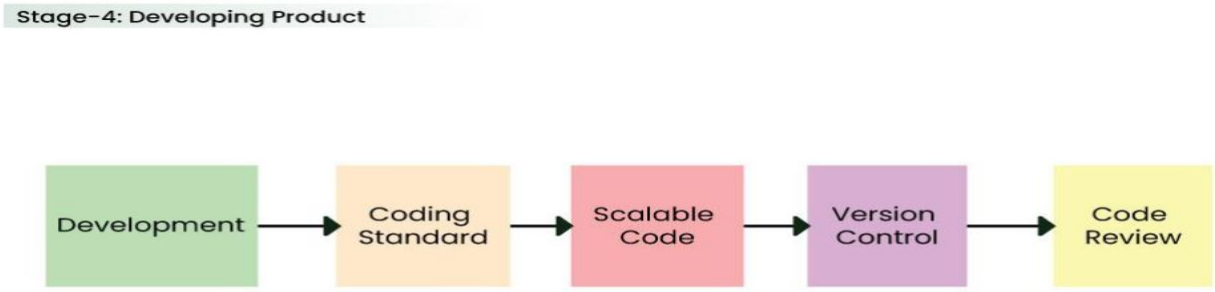


Stage 3: Designing the Product Architecture

- High-Level Design (HLD): Defines the architecture, database design, and relationships between modules.
- Low-Level Design (LLD): Defines the logic of individual components, API interfaces, and database tables.
- Output: Design Document Specification (DDS).
- Key Players: System Architects, Lead Developers.

Stage 4: Building or Developing the Product

- In this stage of SDLC the actual development starts and the product is built. The programming code is generated as per DDS during this stage. If the design is performed in a detailed and organized manner, code generation can be accomplished without much hassle.



Stage 4: Building or Developing the Product

- Activities: Coding, Code Reviews, Unit Testing, and Static Code Analysis.
- Tools: Compilers, Debuggers, IDEs (VS Code, IntelliJ), Version Control (Git).
- Output: Source Code, Executable Software.
- Key Players: Developers (Frontend, Backend, Full Stack).

Stage 5: Testing the Product

- This stage is usually a subset of all the stages as in the modern SDLC models, the testing activities are mostly involved in all the stages of SDLC.
- However, this stage refers to the testing only stage of the product where product defects are reported, tracked, fixed and retested, until the product reaches the quality standards defined in the SRS.

Stage-5: Product Testing and Integration

Stage 5: Testing the Product

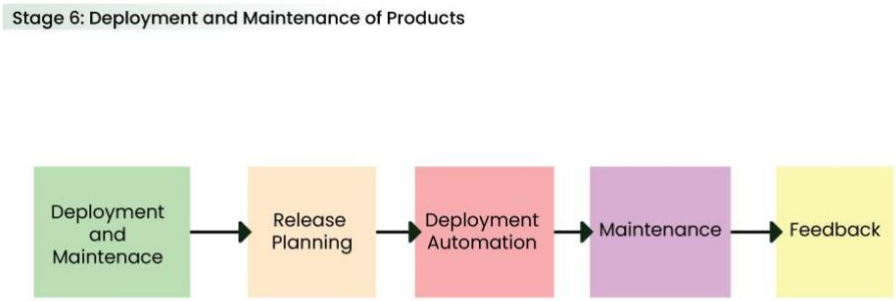
- Types of Testing:
- Unit Testing: Testing individual functions.
- Integration Testing: Ensuring modules work together.
- System Testing: Testing the entire application flow.
- User Acceptance Testing (UAT): Verifying the software meets business needs.
- Output: Bug Reports, Test Cases, Quality Report.
- Key Players: QA Engineers, Testers.

Stage 6: Deployment in the Market and Maintenance

- Once the product is tested and ready to be deployed it is released formally in the appropriate market. Sometimes product deployment happens in stages as per the business strategy of that organization. The product may first be released in a limited segment and tested in the real business environment (UAT- User acceptance testing).
- Then based on the feedback, the product may be released as it is or with suggested enhancements in the targeting market segment. After the product is released in the market, its maintenance is done for the existing customer base.

Stage 6: Deployment in the Market and Maintenance

- Activities: Setting up production environments, deploying code, and smoke testing the live environment.
- Output: Live Application.
- Key Players: DevOps Engineers, Release Managers.



What smoke testing in live/prod usually includes

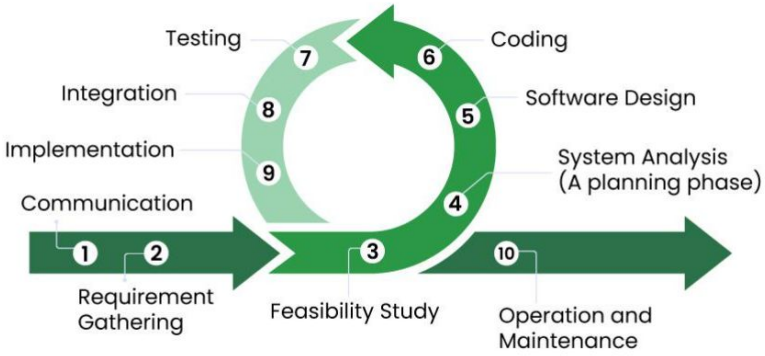
- App loads successfully
- Login works
- Core user flows work (e.g. create order, submit form, payment)
- APIs respond (no 500 errors)
- No major outages or crashes

What is the need for SDLC?

- SDLC is a method, approach, or process that is followed by a software development organization while developing any software.
- SDLC models were introduced to follow a disciplined and systematic method while designing software.
- With the software development life cycle, the process of software design is divided into small parts, which makes the problem more understandable and easier to solve.
- SDLC comprises a detailed description or step-by-step plan for designing, developing, testing, and maintaining the software.

What is Software Development Process?

- The software development process is the approach to developing, and delivering software applications. This process might include improving design and product management by splitting the work into smaller steps or processes.



Need for Software Development Process

- **Meeting Requirements:** The software development process ensures that the final product meets the specified requirements that are needed by the customer.
- **Timely Delivery:** Following a software development process helps in managing project timelines effectively. It checks whether a project is delivered on time or not.
- **Budget Control:** The software development process facilitates budget control by allowing for better estimation and tracking of resources and costs.
- **Quality Assurance:** The process includes testing and quality assurance activities, enabling the identification and resolution of bugs, defects, and usability issues.

Purpose of software development process

- **Organization and Structure:** A development process provides a framework for organizing tasks and activities involved in software development. It defines roles, responsibilities, and communication channels within a development team.
- **Quality Assurance:** The process involves quality assurance and testing stages that ensure that the software fulfills the specifications, is defect-free, and performs as expected.
- **Risk Management**: A well-defined process helps in the early identification of potential risks and provides tools for controlling and reducing those risks.
- **Efficiency and Productivity:** By following a process, development teams can improve efficiency and productivity
- **Communication and Collaboration:** A software development process provides communication and collaboration among team members, stakeholders, and other relevant parties.